Questions by topic: Mineral resources (the lithosphere) - Set 2 (35 marks)

Answer all questions in the spaces provided.

0 1 Table 1 compares the energy used to produce 1 tonne of aluminium from bauxite ore with the energy used to produce 1 tonne from recycled aluminium cans.

Table 1

Source	Energy used / GJ per tonne	CO_2 emitted / tonnes per tonne of aluminium
Bauxite ore (extraction, refining, smelting)	210	12
Recycled cans (collection, sorting, remelting)	12	0.7

0 1 . 1 Calculate the percentage energy saving from recycling aluminium rather than extracting it from ore.

[2 marks]

Answer ☐ %

0 1 . 2 Give **three** advantages of recycling metals other than energy saving.

[3 marks]

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0 1 . 3 Explain **four** difficulties that reduce the effectiveness of recycling schemes.

[4 marks]

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0 1 . 4 Explain what is meant by **Cradle to Cradle** design.

[1 mark]

10

0 2 . 1 As high-grade mineral deposits become depleted, discuss the strategies that could be used to secure future supplies of metals.

[9 marks]

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Extra space

0 2 . 2 State what is meant by the **stock** (resource base) of a mineral.

[1 mark]

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10

0 3 A mining company is assessing a copper deposit beneath farmland.

0 3 . 1 Describe what each of the following survey techniques detects.

[3 marks]

Gravimetry .....

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Magnetometry .....

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Seismic surveys .....

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0 3 . 2 Explain why trial drilling is normally carried out only after the remote sensing surveys.

[2 marks]

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0 3 . 3 Explain how a rise in the world price of copper can turn part of the **resource base** into a **reserve**.

[2 marks]

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0 3 . 4 Explain **four** environmental impacts of extracting the copper by opencast mining.

[4 marks]

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0 3 . 5 Describe how the site could be restored after closure, and explain why restoration is planned before mining begins.

[4 marks]

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END OF QUESTIONS